

Radiation Heat Transfer Tutorial

1 Questions

1. A glass plate 30 cm square is used to view radiation from a furnace. The transmissivity of the glass is 0.5 from 0.2 to $3.5 \mu\text{m}$. The emissivity may be assumed to be 0.3 up to $3.5 \mu\text{m}$ and 0.9 above that. The transmissivity of the glass is zero, except in the range from 0.2 to $3.5 \mu\text{m}$. Assuming that the furnace is a blackbody at 2000°C , calculate the energy absorbed in the glass and the energy transmitted. (transmitted = 58.2 kW, Total absorbed 52.76 kW)
2. Two concentric cylinders having diameters of 10 cm and 20 cm have a length of 20 cm. Calculate the shape factor between the open ends of the cylinders. ($F_{34} = 0.0769$)
3. A truncated cone has top and bottom diameters of 10cm and 20 cm and a height of 10cm. Calculate the shape factor between the top surface and the side and also the shape factor between the side and itself. (bottom -1, top -2, side -3 $F_{23} = 0.52$, $F_{33} = 0.397$)
4. Two parallel plates 0.5m by 1.0m are spaced 0.5 m apart. One plate is maintained at 1000°C and the other at 500°C . The emissivities of the plates are 0.2 and 0.5 respectively. The plates are located in a very large room, but only the plates facing each other are to be considered in the analysis. Find the heat transfer to each plate and to the room. (Heat lost by plate 1- 14.425 kW, Plate 2.594 kW, Heat received by room 17.02 kW)
5. Two rectangles 50 cm by 50 cm are placed perpendicularly with a common edge. One surface has 1000 K, with emissivity 0.6, while the other surface is insulated and in radiant balance with a large surrounding room at 300 K. Determine the temperature of the insulated surface and heat lost by the hot surface. (Heat 8.3 kW, Temp 599.4 K).
6. Two very large parallel planes with emissivities 0.3 and 0.8 exchange heat. Find the percentage reduction in heat transfer when a polished aluminum radiation shield ($\epsilon = 0.04$) is placed between them. (93.2 %)

7. A surface transmits 80 % of the radiation incident on it in the range 0.3 to 1.0 μ m and it is opaque for radiation of any other wave length . Estimate the heat flux passing through the surface if it is receiving radiation from a black surface at 1500K.
8. A brick wall of emissivity 0.8, 6m wide $\times$ 4m high is located at a distance of 4m from an opening of size 20 cm $\times$ 20 cm in a furnace wall. The centre line of the opening lies 1m lower and 1m left of the centre of the wall. If the furnace operates at a temperature of 1523 °C and the wall temperature is 37 °C, Calculate the radiant heat exchange between the opening and the wall.
9. Calculate the following for an industrial furnace in the form of a black body and emitting radiation at 2500 °C (a) Monochromatic emissive power at 1.2 μ m (b) Wave length at which the emission is maximum. (c) Maximum emissive power. (d) Total emissive power (e) Total emissive power of the furnace if it is assumed as a real surface with emissivity 0.9
10. Two large parallel planes with emissivities 0.3 and 0.8 exchange heat. Find the percentage reduction when a polished aluminum shield of emissivity 0.04 is placed between them. Use the method of electrical analogy.